

Autonomous ML Algorithm for Satellite Image Optimization

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Industry Partner: Beyond Gravity

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1 Existing Methods and Research

Current satellite imaging relies on pre-scheduled captures or ground-based analysis, using tools like NASA's Earth Observing System for cloud detection via spectral analysis and basic ML models (e.g., CNNs for classification). Research includes ESA's AI4EO for on-board autonomy, with lightweight segmentation (U-Net) and orbit propagation libraries (Orekit). Thangavel et al. (2023) demonstrate TASO in DSS using optical sensors for wildfire detection with 1D CNNs on hyperspectral data, achieving high system-wide access (75-98%) via reconfiguration. Thangavel et al. (2023) also propose a DSS constellation of formations for multi-baseline AT-InSAR in MDA, using autonomous orbit control with low-thrust actuation for formation maintenance and reconfiguration. Limitations: High latency from ground dependency, inefficient handling of dynamic weather, and resource-heavy models not optimized for satellite constraints.

2 Proposed Improvements

We aim to enhance autonomy by enabling on-board real-time image analysis and adaptive maneuvering, reducing ground intervention and improving capture efficiency under variable conditions. This shifts from reactive to predictive operations via probabilistic modeling, potentially evolving into a reinforcement learning (RL) framework (e.g., actor-critic ensemble for imaging and decision-making)—extending Thangavel et al. by focusing on general cloud/fog avoidance using front-camera images, without hyperspectral or SAR specificity.

3 Methodology and Tools

Improvements via probabilistic approach:

- **Approach Target and Analyze:** On approaching the imaging target, analyze front-camera images for weather conditions (clear/cloudy/foggy) using lightweight ML (U-Net for segmentation, CNN for classification) with TensorFlow Lite; map to ground coordinates via online datasets (NASA/ESA).

- **Probabilistic Success Assignment:** Assign success probabilities for imaging angles/strategies based on mission requests (critic model evaluates likelihood).
- **Decision and Maneuvering:** If desired success threshold is met, release actor to command AOCS for optimal pointing/maneuvers (direction, angle, speed, times) using Poliastro for orbital propagation and parametric AOCS.
- **Window Prediction:** Compute future imaging opportunities; support ground-injected areas.

Prototype in Python simulation, validated with public data. Outcomes: Feasible proof-of-concept model, report on scalability.

4 References

Thangavel, K. et al. (2023). Trusted Autonomous Operations of Distributed Satellite Systems Using Optical Sensors. *Sensors*, 23(6), 3344; Thangavel, K. et al. (2023). A Distributed Satellite System for Multibaseline AT-InSAR: Constellation of Formations for Maritime Domain Awareness Using Autonomous Orbit Control. *Aerospace*, 10(2), 176; ESA AI4EO; U-Net papers; Orekit documentation.